

WAKE ISLAND, United States Minor Outlying Islands

WMO: 912460

Lat: 19.283N

Lon: 166.650E

Elev: 23

StdP: 14.68

Time Zone: 12.00 (E12)

Period: 94-19

WBAN: 41606

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	72.3	72.9	57.4	70.4	74.7	60.0	77.3	75.9	32.3	73.9	28.8	75.4	15.4	50	0.569

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.3	90.8	79.8	90.1	79.7	89.1	79.4	83.4	85.6	82.3	84.6	81.8	84.1	11.2	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
82.4	169.0	83.9	81.8	165.8	83.2	81.3	162.7	82.8	47.2	86.2	46.5	83.1	45.2	85.2	90.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.9	24.9	23.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.6	78.6	78.2	78.6	80.3	81.9	83.8	83.9	84.3	84.2	83.3	81.8	80.1
	DBStd	2.83	1.83	1.97	1.55	1.74	1.82	1.57	1.78	1.87	1.65	1.75	1.56	1.63
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11529	888	789	886	908	987	1013	1052	1062	1026	1032	953	933
	CDD65	6054	423	369	421	458	522	563	587	597	576	567	503	468
	CDH74	63038	3156	2568	3086	4204	5531	6679	7085	7358	7073	6702	5312	4284
	CDH80	18819	387	309	405	881	1551	2447	2712	2979	2816	2335	1274	723
(14) Wind	WSAvg	13.1	12.8	13.6	15.0	14.4	12.1	11.7	12.3	10.9	11.2	13.4	16.0	13.4
(15) Precipitation	PrecAvg	29.7	0.9	1.3	1.6	1.7	1.4	2.0	3.3	4.0	3.8	3.6	2.3	1.3
	PrecMax	48.6	4.3	6.5	6.9	6.2	5.6	5.4	10.7	14.2	16.1	11.7	5.8	4.8
	PrecMin	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	12.5	0.9	1.5	1.9	1.5	1.3	1.6	3.2	3.7	3.2	3.0	1.7	1.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	87.1	84.5	85.0	88.3	91.4	93.1	92.7	91.6	91.1	90.4	88.2	86.4
		MCWB	76.1	75.8	74.9	76.7	77.7	79.8	79.7	81.3	81.3	79.6	77.6	76.6
	2%	DB	84.2	83.9	84.0	86.4	88.4	90.2	90.3	90.3	90.1	88.7	86.5	85.2
		MCWB	74.8	75.3	74.4	76.2	77.8	79.1	79.5	80.5	80.5	79.3	77.3	76.4
	5%	DB	83.0	82.5	82.9	85.5	87.2	88.5	89.0	89.5	89.1	88.0	85.9	84.2
		MCWB	74.3	74.2	73.9	76.1	77.4	78.3	79.4	80.0	79.9	79.1	77.1	75.9
	10%	DB	82.1	81.9	82.1	84.1	85.9	87.8	88.1	88.3	88.1	87.0	84.8	83.6
		MCWB	73.8	73.8	73.5	75.5	77.1	78.0	79.2	79.7	79.7	78.7	76.8	75.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.3	78.3	78.1	79.2	81.4	82.5	83.4	87.2	85.4	83.4	82.4	79.4
		MCDB	81.7	82.1	80.5	82.3	83.2	84.7	86.0	88.5	86.6	85.3	82.8	81.7
	2%	WB	77.1	77.2	76.8	78.1	80.2	81.3	82.2	83.4	83.4	82.2	81.2	78.4
		MCDB	81.5	81.2	80.2	82.6	83.1	84.3	84.8	86.0	85.6	84.3	82.1	81.7
	5%	WB	76.2	76.2	76.0	77.3	79.2	80.4	81.4	82.0	82.3	81.4	79.7	77.7
		MCDB	81.0	80.7	80.2	82.6	83.2	84.6	84.2	84.5	84.6	83.6	82.4	81.8
	10%	WB	75.4	75.4	75.2	76.6	78.3	79.5	80.8	81.2	81.5	80.7	78.6	77.0
		MCDB	80.5	80.1	79.9	82.0	83.4	84.8	84.5	84.2	83.9	83.7	82.4	81.5
(35) Mean Daily Temperature Range	5% DB	MDBR	7.1	7.3	7.9	8.3	8.2	8.3	8.3	8.3	8.1	7.6	6.8	6.9
		MCDBR	8.0	8.0	8.5	9.5	9.7	9.6	9.4	9.3	9.1	8.6	7.6	7.8
		MCWBR	4.0	4.1	4.0	3.9	3.8	3.9	4.0	4.2	4.4	4.0	4.0	4.0
	5% WB	MCDBR	7.6	7.6	8.3	8.5	8.4	9.1	9.1	8.6	8.4	8.0	7.4	7.1
		MCWBR	4.2	4.1	4.5	4.1	4.3	4.7	4.7	5.2	5.0	4.6	4.5	4.0
(40) Clear-Sky Solar Irradiance	taub		0.370	0.375	0.396	0.415	0.406	0.399	0.416	0.415	0.408	0.417	0.402	0.381
	taud		2.582	2.558	2.506	2.474	2.523	2.556	2.516	2.518	2.545	2.495	2.504	2.547
	Ebn at Noon		289	293	290	283	282	281	277	279	282	276	276	281
	Edh at Noon		30	32	35	36	34	33	34	34	33	33	32	30
(44) All-Sky Solar Radiation	RadAvg		1512	1738	1984	2159	2228	2246	2050	1909	1834	1704	1559	1464
	RadStd		60	95	60	104	75	102	155	192	136	87	63	48

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page